THE OPEN UNIVERSITY OF SRI LANKA
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING
BACHELOR OF TECHNOLOGY - LEVEL 6

EEX6535 – COMPILER DESIGN

CONTINUOUS ASSESSMENT TEST #2 - 2019/ 2020 **CLOSED BOOK TEST**

Date: 30 January 2020 **Time: 0900 – 1015 hrs**

Answer all the questions. State your assumptions (if any) clearly.

[Q1]. Consider the grammar rules are given below (VendingMachine, STOCK, SALES, ITEM, SALE are non-terminals and others are terminals).

VendingMachine → stock STOCK sales SALES
STOCK → STOCK ITEM | ε
ITEM → item price qty
SALES → SALES SALE | ε
SALE → item price

(a) Derive the string: ***stock item price qty sales item price*** **[10 Marks]**

(b) Define the Chomsky Normal Form (CNF) for CFGs. **[10 Marks]**

(c) Convert the given grammar into CNF. **[40 Marks]**

(d) Derive the above string in (a) using new grammar in (c) **[10 Marks]**

[Q2].

(a) Briefly explain the LR parsing by giving examples. **[10 Marks]**

(b) Compare and contrast the top down vs. bottom up parsers. **[20 Marks]**